

MESQUITE, TX, USA

WMO: 720291

Lat: 32.747N

Lon: 96.531W

Elev: 447

StdP: 14.46

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53970

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.5	27.6	10.3	9.4	32.0	14.8	11.8	35.6	26.3	56.7	24.2	55.1	7.2	330	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.3	100.1	73.3	98.8	73.3	95.9	73.4	78.2	89.5	77.3	88.8	76.5	88.0	7.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	134.5	82.5	73.5	126.6	80.8	73.1	124.8	80.7	42.0	87.9	41.0	88.7	40.1	88.1	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.5	20.1	18.0	DB	18.2	103.9	4.9	2.7	14.6	105.9	11.8	107.5	9.0	109.0	5.4	110.9	
			WB	16.3	79.9	4.4	1.7	13.2	81.1	10.6	82.0	8.2	83.0	5.0	84.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	46.2	49.3	58.0	64.7	72.6	80.9	83.4	84.2	77.0	66.3	57.0	46.9
	DBStd	15.57	9.83	10.40	9.04	7.73	6.68	3.63	3.58	4.71	5.75	7.87	8.88	9.83
	HDD50	568	191	126	32	4	0	0	0	0	0	1	32	182
	HDD65	2315	587	444	251	96	21	0	0	0	1	80	267	568
	CDD50	6269	73	107	279	445	699	927	1035	1061	810	506	243	84
	CDD65	2544	4	5	33	88	255	477	570	596	361	121	28	6
	CDH74	28149	21	47	196	692	2270	5314	6885	7570	3821	1099	197	37
	CDH80	13693	2	2	20	167	799	2600	3683	4252	1815	319	31	3
	WSAvg	7.5	7.8	8.8	9.8	9.8	8.0	7.2	5.9	5.5	5.6	6.4	7.6	7.7
	PrecAvg	38.4	2.7	2.6	3.5	3.7	4.8	3.6	2.0	2.2	2.8	4.3	3.1	3.0
(15) Precipitation	PrecMax	66.3	6.3	7.4	7.0	7.3	15.2	10.7	4.9	5.1	9.3	11.7	10.2	8.7
	PrecMin	20.9	0.2	0.2	0.1	1.2	0.6	0.7	0.0	0.0	0.5	0.3	0.5	0.2
	PrecStd	10.0	1.6	1.8	1.9	1.8	2.8	2.6	1.5	1.4	2.2	3.0	2.3	2.0
	PrecStd	10.0	1.6	1.8	1.9	1.8	2.8	2.6	1.5	1.4	2.2	3.0	2.3	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.8	80.5	82.1	89.6	92.8	98.9	101.6	104.4	99.4	90.8	83.8	79.1
		MCWB	60.8	62.4	65.0	70.9	72.0	72.5	73.9	73.5	70.8	71.2	66.8	62.6
	2%	DB	72.2	73.5	79.1	84.0	90.3	96.6	99.3	101.7	96.3	87.5	79.4	72.3
		MCWB	59.4	58.7	63.9	68.9	72.0	72.5	73.9	73.1	71.3	71.5	64.9	61.9
	5%	DB	68.1	70.4	75.4	81.2	88.0	94.6	97.1	99.6	92.8	82.5	75.1	68.1
		MCWB	59.9	59.7	63.1	66.9	72.2	73.1	73.8	73.3	71.0	67.4	64.2	59.4
	10%	DB	63.7	66.1	72.6	78.6	84.3	91.2	94.8	97.1	90.3	80.9	72.1	63.7
		MCWB	56.0	57.6	61.5	65.6	70.8	73.7	74.2	73.8	70.5	66.6	62.8	56.5
	0.4%	WB	67.4	66.3	69.5	73.4	77.2	79.3	79.3	79.8	77.5	76.0	71.0	67.3
		MCDB	69.1	74.5	78.1	84.6	86.2	89.5	89.7	90.4	88.2	83.5	77.9	73.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	64.4	64.2	67.2	71.2	75.2	77.7	77.9	78.1	76.3	74.2	68.2	64.4
		MCDB	68.3	70.4	74.6	80.3	84.9	87.7	89.4	90.5	86.9	81.6	75.2	69.5
	5%	WB	61.7	62.0	65.8	69.4	73.9	76.6	77.1	77.1	74.9	72.3	66.2	62.0
		MCDB	66.6	67.7	72.9	77.1	83.5	87.1	88.8	89.9	85.3	79.1	73.1	66.2
	10%	WB	56.7	59.2	64.0	68.0	72.5	75.5	76.2	76.0	73.5	70.4	64.0	57.9
		MCDB	63.2	65.1	69.9	75.8	81.9	86.5	87.8	88.9	83.6	76.8	69.4	63.4
(35) Mean Daily Temperature Range	5% DB	MDBR	20.7	20.6	21.2	21.5	19.9	20.1	20.9	22.3	22.0	22.8	21.6	20.7
		MCDBR	27.7	26.0	23.0	22.9	22.5	23.4	23.9	25.8	25.7	24.9	24.4	26.8
		MCWBR	17.4	16.1	12.7	11.3	9.1	7.0	6.0	6.2	8.7	11.9	14.0	17.4
	5% WB	MCDBR	21.7	21.9	19.7	19.2	18.6	19.0	19.7	21.4	20.2	20.2	20.2	21.4
		MCWBR	16.8	15.7	12.7	11.5	9.3	7.6	6.4	6.9	8.1	11.4	14.0	16.6
(40) Clear-Sky Solar Irradiance	taub		0.317	0.333	0.360	0.398	0.429	0.442	0.445	0.447	0.402	0.360	0.341	0.322
	taud		2.496	2.462	2.403	2.316	2.289	2.286	2.317	2.295	2.370	2.473	2.470	2.486
	Ebn at Noon		286	292	291	283	274	269	267	266	274	279	275	278
	Edh at Noon		28	31	36	41	43	43	41	41	37	31	28	26
	RadAvg		896	1037	1387	1692	1827	2058	2105	1938	1630	1280	954	791
(45) All-Sky Solar Radiation	RadStd		87	164	112	122	155	195	143	129	141	190	121	102

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page